

Per-Tier and End-to-End Metrics (Held-Out Test)

Component	Task	N	Accuracy	Precision	Recall	F1	Extra
Tier-1 Gate	Binary (escalate)	14219	1.0000	1.0000	1.0000	1.0000	—
Tier-2 CNN-GRU	6-class	14219	0.9805	0.9859	0.9850	0.9849	—
Tier-3 One-Class	Anomaly	1896	0.9953	0.9918	1.0000	0.9959	AUC=1.0000
Full Cascade	E2E security	14219	0.9994	0.9993	1.0000	0.9997	FPR=1.11%

Macro precision/recall/F1 for Tier-2. E2E: malicious = BLOCK or FLAG.